Sahaj Bhadja

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Summary

Backend Software Engineer with 2+ years building **Python** and Java REST microservices, **third-party API** integrations, data pipelines on PostgreSQL, **payment** and fraud-aware workflows, and AI agent systems shipped with Docker and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **data pipelines** that ingest, validate, and reconcile records across PostgreSQL, SQL Server, and cloud storage, sustaining 99%+ reliability over 5K+ monthly jobs through CI/CD-backed releases for 500+ users.
- Designed multi-tenant **PostgreSQL** data models with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms, and tripled async AI job throughput across 1K+ jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Integrated OpenAI, Anthropic, and Qwen APIs behind one **switchable provider layer** with dynamic model selection, OCR fallbacks, and timeout handling, serving 10K+ daily requests for 150+ instructors on AWS.
- Built secure **document ingestion** for PDF, DOCX, and ZIP files with multithreaded processing of 1K+ files per run, and debugged production failures to cut feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built **microservices** and large-scale API integrations with object-oriented Factory and Singleton patterns, improving response times 30% through pagination, field filtering, and caching for low-bandwidth users across distributed clients.
- Redesigned **ETL** on AWS Glue and S3 that transformed JSON, CSV, and Parquet data from multiple sources, boosting pipeline efficiency 68%, backed by 20+ automated tests in Jenkins.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **CRM** records platform with React TypeScript screens and Node.js microservices for customer records, interaction history, billing, and role-based access, syncing 2M+ records in real time.
- Added **monitoring** with Sentry, Azure Monitor, and Kibana, and shipped through Jenkins CI/CD with Docker packaging, automated tests, and approval-gated releases, cutting deployment time 50% across staging and production.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built PCI-compliant **payment** infrastructure for cards, wallets, and PayPal with **fraud** detection, refund, dispute, and chargeback handling through idempotent webhooks, cutting manual handling about 70% across 50K+ booking records.
- Built Flask booking and coupon APIs with eligibility checks and transactional locking to prevent double bookings, keeping responses under 200ms across 50K+ records, and presented the architecture to the client.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL
Backend: FastAPI, Flask, Spring Boot, Node.js, REST APIs, Microservices, Webhooks, Third-Party APIs
Data: PostgreSQL, MySQL, SQL Server, MongoDB, Data Modeling, ETL Pipelines, AWS Glue, Spark
AI & Frontend: LLM APIs, OCR, Embeddings, Prompt Engineering, React, Debugging, Agile, Linux
Infra & Practices: Docker, Kubernetes, AWS, CI/CD, Jenkins, GitHub Actions, Git, Sentry, Kibana, PyTest, Code Reviews

Projects

KYB Verification Orchestrator | *Python, FastAPI, PostgreSQL, SQS, Docker*

- Designed a **FastAPI** service that runs business verification across several vendors in parallel, maps each vendor's response into one schema, retries failed calls through SQS, and opens **review cases** for mismatches.

AI Sanctions Screening Agent | *Python, LLM APIs, Elasticsearch, PostgreSQL*

- Designed an **AI agent** that fuzzy-matches applicants against sanctions lists in Elasticsearch, summarizes adverse-media hits with sources, and recommends clear, escalate, or reject with reasons for **analyst review**.

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, Kubernetes*

- Built a **fintech** platform with Plaid bank linking, a token-based firewall, and real-time transaction updates, supporting 50,000+ users and 1M+ transactions with Kubernetes CI/CD deployments and secure access.